

Amna Munawar

Senior Data Engineer

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SUMMARY

Senior Data Engineer with end to end ownership of an AWS data lakehouse platform processing **10M+** records daily on Kafka, Spark, Airflow, Trino and dbt, cutting platform compute and storage costs **25 to 35%**. Ships AI powered data products including a live NL to SQL agent platform, and acts as first responder for production data platform incidents.

SKILLS

Data engineering & storage: SQL, Python, PySpark, Kafka, Debezium (CDC), Trino, dbt, Airflow, Apache Iceberg, Hive Metastore, DuckDB, PostgreSQL, MySQL, ClickHouse, TimescaleDB, MongoDB / DocumentDB, data modeling, SCD Type 2, schema evolution, Apache Ranger

Cloud & Infrastructure: AWS (S3, RDS, EKS, EMR, Glue, Lambda, Athena, Redshift, DMS, VPC PrivateLink, IAM), Kubernetes, CNPG, Docker, Helm, Terraform, ArgoCD, CI/CD (GitHub Actions), Databricks / Delta Lake, GCP BigQuery, KEDA, Carpenter

Observability & analytics: Grafana, Percona, JMX, Metabase, Superset, Looker, LightDash, Tableau, OpenMetadata (cataloging and lineage), dbt tests, data contracts, anomaly detection

AI & machine learning: RAG, LangChain, LangGraph, CrewAI, multi agent orchestration, pgvector, vector search, MLflow, scikit learn, Isolation Forest, ALS recommendation (matrix factorization), OpenAI, Google Gemini, prompt engineering

EXPERIENCE

Bazaar Technologies, Karachi, Pakistan

Senior Data Engineer II

July 2025 – Present

- Architected the end to end data lakehouse platform (Spark, Kafka, Airflow, Trino, dbt, S3, Iceberg) processing **10M+** records daily, cutting time to insight by **70%** across 10 business teams.
- Engineered real time streaming pipelines on TimescaleDB hypertables and continuous aggregates on Kubernetes (CNPG), reducing the data latency SLA from 20 minutes to under 1 minute.
- Deployed an AI agent powered NL to SQL platform using RAG over a medallion architecture, enabling self service analytics and cutting analyst workload by **60%**.
- Built Python hot and cold tiering pipelines archiving **65M+** MongoDB documents to Apache Parquet, cutting storage costs by **30%**.
- Acted as first responder for production data platform incidents across Trino, Airflow, Hive Metastore and Superset, root causing outages end to end and shipping the permanent fix.
- Implemented Hive Metastore as a centralized schema registry with data contracts across Trino and dbt, cutting schema inconsistencies by **70%**.

Senior Data Engineer I

January 2024 – June 2025

- Optimized Trino cluster performance via KEDA autoscaling, Kubernetes resource tuning and query routing, cutting infrastructure costs by **25%** and boosting throughput by **40%**.
- Cut S3 LIST request cost by **82%**, from **\$100/month to \$18/month**, across two Trino clusters by tuning the Hive file status cache, using JMX instrumentation to attribute request cost per model.
- Improved Airflow reliability by refactoring resource heavy DAGs with modular Python and idempotent backfills, cutting SLA breaches by **80%** and raising pipeline SLO to **99%**.

Software Engineer II

January 2023 – December 2023

- Migrated legacy Spark workloads to a Trino and dbt medallion lakehouse with optimized ELT transformations, cutting pipeline runtime by **45%** and compute costs by **35%**.
- Built a data quality and discovery platform using OpenMetadata, dbt validation tests and Metabase alerting, reducing data quality incidents by **95%**.
- Built an AWS serverless ETL path (S3 raw to Glue and Lambda transforms to Redshift Serverless gold KPIs to Athena to Metabase), cutting Athena data scanned by **80%**.

Software Engineer I

July 2022 – December 2022

- Replaced full RDS snapshot backups with a DuckDB powered partition level Parquet pipeline on S3 via Airflow, cutting storage costs by **60%** and restore time from hours to minutes.

EDUCATION

Mohammad Ali Jinnah University, M.S. Data Science, 4/4 (2025 to 2027)

FAST University, B.S. Computer Science, 3.6/4 (2018 to 2022)